Research Statement

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AI has made programming more accessible and efficient, but it has also introduced new challenges related to software reliability. My overarching research goal is to make automated programming more trustworthy by developing AI systems that generate more robust and correct code, alongside techniques that can rigorously verify and test this code. I believe AI must not only generate correct answers but also reason about their *soundness* and derive *general principles* from experience. To achieve this, I consider understanding the following aspects essential:

1. **Language models capability to reason about *program semantics***: Language models can generate syntactically correct code but often fail at logical reasoning tasks requiring systematic proof construction. How does a model's training regime shape its capacity for formal reasoning versus pattern matching?

2. **Experiential learning in program synthesis**: Synthesis systems can distill reusable knowledge from past attempts, akin to library learning. Yet learned strategies often remain task-specific. How do the granularity of stored knowledge and retrieval mechanisms shape the transferability of problem-solving patterns?

3. **Verification-guided synthesis**: Current code generation approaches treat verification as a post-hoc validation step, requiring separate proof construction after synthesis. How can we design training procedures where correctness constraints actively shape the generation process, making proof obligations explicit during synthesis?

My interest in this field has been the result of my past research experience in varied subjects, which I describe in the following section.

Research on formal reasoning with LLMs: My research at the intersection of AI and formal methods started during my pre-doctoral fellowship at Microsoft PROSE, where I worked on translating natural language intent to formal specifications. Neural models often generate plausible-looking specifications that subtly diverge from user intent, undermining formal soundness guarantees. While exploring Logic-LM, a technique using theorem prover feedback to refine formal statements, I found out that prover feedback fixed syntax errors but led models toward formally valid yet semantically misaligned statements. This insight motivated me to develop a neurosymbolic scoring system combining LLM perplexity (for semantic alignment) with prover errors (for formal validity), reducing both syntactic errors and semantic drift while requiring fewer refinement iterations. This work was published at the **Natural Language Reasoning workshop at ACL 2024** [Kirtania et al. 2024]. **In my PhD, I would like to work on neurosymbolic techniques that can repair formal statements and proofs with *soundness* and *correctness* guarantees.**

Architectures like Logic-LM were severely limited to post-hoc improvements of the LLMs generations. This motivated me to ask a deeper question: *"What internal mechanisms enable LLMs to perform formal reasoning, and can we enhance these mechanisms directly?"*. Through a mechanistic analysis of how LLMs interpret proofs in Lean4, I discovered that LLMs use different internal mechanisms to reason about formal and informal statements. This motivated me to explore the use of technique of *activation steering*: a technique to influence a model's internal mechanisms by creating a vector of the activations of desired behavior. I explored the extent to which activation steering can impact the theorem proving capabilities for LLMs. Use activation steering substantially improved model performance on standard benchmarks. My analysis revealed that steering effects concentrate in the model's latter layers, suggesting that proof construction primarily occurs in these deeper computational stages. Remarkably, vectors learned from Lean4 proofs demonstrated improvement in performance in Rocq Prover as well, demonstrating their capture of fundamental reasoning patterns rather than system-specific syntax.

This progression from post-hoc improvements to internal guidance represents a notable reflection point in neural theorem proving. By understanding and manipulating internal representations, we can create AI systems that genuinely understand formal mathematics rather than merely mimicking it. This work, is **under submission at top ML conference** [Kirtania and Iyer 2025]. Alongside this, I also contributed to a formal math benchmark in Lean where, I focused on formalizing Olympiad problems in mathematics and a VS code plugin for AI assisted formalization [Biyani et al. 2025]. This work got accepted as an **oral presentation at AAAI Formal Methods for AI workshop** and is **under submission at a top ML conference**.

Building on the mechanistic insights and quantitative analysis, I am currently collaborating with **Prof. Emily First** to investigate representation of proof strategies in foundational models and their fine-tuned theorem proving variants. **As part of my PhD, I want to research on data driven techniques to make verified code generation simpler and automated theorem proving more efficient.**

Research in LLM Agents for Software Systems: My work on formal reasoning focused on how modes construct proofs. However, automated programming is not only about logical correctness but also about learning correct behaviors from failure. This question led me to explore experiential learning in language agents. I addressed their inability to learn from experience, as agents repeat mistakes by approaching each task anew, unlike humans who internalize these patterns from experience.

I developed two solutions. First, *MetaReflection* [Gupta* et al. 2024], inspired by counter-example guided inductive synthesis (CEGIS). It learns optimal prompt-level instructions from ground-truth counterexamples, enabling unsupervised task improvement. I applied this to vulnerability detection in Infrastructure-as-Code tools augmented with static analyzers to ensure correctness. This work was presented at **EMNLP 2024** and later on adapted by Github for secret key detection [Github-Engineering 2025].

Recognizing that CEGIS-synthesized programs are more deterministic than natural language, leading to semantically similar rather than optimal instructions, I developed *BREW*. This extends experiential learning to structured knowledge bases (KBs) using an *Expand-and-Gather MCTS algorithm* that treats memory optimization as search rather than gradient updates. On benchmarks like *OS-World*, $\tau - Bench$ and *SpreadSheetBench*, BREW improved both performance and efficiency with just 25 training examples. This work is **under submission at top ML conference** [**Kirtania** et al. 2025]. **In my PhD, I would like to make neurosymbolic techniques to make compositional abstractions that are scalable and efficient.**

While Knowledge base (KB) generation helps when KBs are absent, existing KBs often require refinement rather than replacement. Agentic systems fail when KBs contain errors. This issue is critical in dynamic software engineering environments where documentation lags behind rapid changes in private libraries. I worked on a novel KB refinement technique using agents, *StackFeed* [**Kirtania** et al. 2025]. In *StackFeed* I formalize KB refinement as a Multi-Actor Centralized Critic problem: multiple LLM actors propose edits while a symbolic critic uses expert feedback to assign attainable sub-tasks. I evaluated *StackFeed* on scenraios like Java version migration and perturbed libraries where it systematically improved KB coherence and completeness without manual intervention. This work was presented at **EMNLP 2025** and **NeurIPS DL4C workshop**. **During my PhD, I want to make symbolic tools to edit model and their artifacts reliably to ensure correctness and completeness.**

At the **University ABC**, I would like to work with Prof. **X** and Prof. **Y**. **With Prof. X**, I hope to explore **«broad research overlap»**. Building upon Prof. X's prior work on **«referred to two core papers»**, I am curious to explore **«how can we build upon the two papers»**. **With Prof. Y**, I am interested **«broad research overlap»**, an area where my prior experience **«talked about particular experiences and how they match Prof Y's area of interest»**.

Pursuing a PhD at University of Michigan, Ann Arbor would be the most fulfilling and intellectually rewarding endeavor of my career. I find exploratory research deeply exciting because of its potential to transform how we think and build. After my PhD, I aspire to remain in academia, where the open-ended nature of university research fosters creativity and interdisciplinary collaboration. I am equally enthusiastic about mentoring and teaching, as I believe guiding and learning from students is one of the most valuable aspects of academic life.

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